

Why Zero-Point Vacuum Energy Is Not a Cosmological Prediction in Quantum Field Theory

An Invariance-Based Analysis of the Cosmological Constant Problem

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Abstract

The cosmological constant problem is often introduced in quantum field theory by comparing the observed large-scale curvature term with a cutoff-regulated sum of zero-point energies. This comparison treats the absolute zero-point energy of a fixed vacuum sector as a standalone cosmological prediction. This paper argues for a narrower and more defensible conclusion: that particular comparison is not a prediction failure, because the quantity being compared is not a renormalization-invariant standalone prediction of QFT. The stress-energy tensor is a composite operator whose finite definition is ambiguous by local conserved counterterms. The metric-proportional ambiguity shifts the homogeneous vacuum-energy coefficient while preserving locality, covariance, conservation, and the empirical normalization of non-vacuum matter observables. We formulate the obstruction as a classification under the homogeneous metric-proportional finite renormalization: within the fixed-background homogeneous vacuum sector, invariant quantities are either boundary-, phase-, or configuration-conditioned differences, where the common additive ambiguity cancels, or the gravitationally renormalized effective cosmological constant. In semiclassical gravity, no curvature observation separates a shift of matter-sector vacuum energy from a compensating shift of the cosmological constant. Casimir forces and phase-transition energy differences are not counterexamples, because they are relational response or transition quantities rather than measurements of an absolute vacuum offset. The result does not explain the small observed value of the renormalized cosmological constant and does not remove the associated naturalness or matching problem. It removes only the claim that a regulator-dependent zero-point sum is itself a cosmological prediction of QFT.

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1 Introduction

The standard presentation of the cosmological constant problem begins with the zero-point energy of quantum fields. For one relativistic bosonic degree of freedom, a cutoff calculation gives, schematically,

$$\rho_{\text{zp}}(\Lambda_{\text{UV}}) = \frac{1}{2} \int^{\Lambda_{\text{UV}}} \frac{d^3k}{(2\pi)^3} \sqrt{k^2 + m^2} \sim \frac{\Lambda_{\text{UV}}^4}{16\pi^2} + \frac{m^2 \Lambda_{\text{UV}}^2}{16\pi^2} + \dots \quad (1)$$

If the cutoff is identified with a particle-physics or Planckian scale, the inferred density is enormously larger than the density associated with the observed acceleration of the universe. This mismatch is a central motif in the literature on the cosmological constant problem [1–4].

This paper addresses a specific question that is logically prior to any proposed cancellation or selection mechanism:

Does the absolute zero-point energy appearing in Eq. (1) constitute a standalone cosmological prediction of quantum field theory?

The answer defended here is no. The regulated expression is an intermediate contribution to a composite operator. Its finite value is convention-dependent. In a renormalized treatment, the homogeneous metric-proportional part of the stress tensor can be shifted by a finite local counterterm. In semiclassical gravity the same shift is exactly absorbed into the cosmological constant. The split between “matter vacuum energy” and “bare cosmological constant” is therefore not separately identifiable.

The claim is intentionally limited. The argument does not deny the observed acceleration of the universe. It does not deny that a gravitational effective action contains a renormalized cosmological term. It does not deny the physical relevance of energy differences between dynamically connected vacua, phases, or boundary configurations. It also does not solve the technical-naturalness problem associated with the small measured value of the effective cosmological constant. The target is narrower: the direct comparison between a regulator-dependent zero-point sum and a measured cosmological parameter is not a comparison between a QFT prediction and observation.

1.1 Relation to existing work

Several ingredients of the argument are standard. Weinberg’s review frames the smallness of the cosmological constant as a deep puzzle for particle physics and cosmology [1]. Curved-spacetime QFT shows that the renormalized stress-energy tensor is defined only up to finite local conserved curvature terms [5–11]. Effective-field-theory treatments emphasize that the cosmological constant is a relevant gravitational coupling and that its radiative stability is a naturalness issue [4, 12]. Foundations-oriented analyses have argued explicitly that the vacuum expectation value of energy

density fails requirements for being a QFT prediction, and that the cosmological constant problem exposes limitations in applying ordinary EFT reasoning to gravity [13, 14].

The present contribution is correspondingly modest but precise. It does not claim priority for the elementary facts above. It repackages them as a scope-controlled invariance classification. The contribution of this paper is not the observation that bare vacuum energy is scheme-dependent; it is a classification showing that the zero-point comparison does not define a valid prediction. The result isolates the exact point at which the standard many-orders-of-magnitude argument leaves the space of invariant predictions. Once the homogeneous metric-proportional finite renormalization is included in the equivalence relation, the fixed-vacuum absolute zero-point coefficient has no nonconstant invariant content. The remaining invariant structures are relational differences or the total renormalized gravitational coefficient. This classification is useful because it identifies the exact boundary between two claims that are often conflated:

- (i) the invalid claim that a cutoff zero-point sum is a standalone prediction of the observed cosmological constant;
- (ii) the valid and still difficult question of why the renormalized effective cosmological constant has its measured small value.

1.2 Main result and scope discipline

The result should be read as a restricted no-prediction theorem, not as a universal theorem about every stress-tensor observable and not as a solution to the observed smallness of dark energy. Its domain is the homogeneous metric-proportional part of the vacuum stress tensor that enters the usual zero-point argument. Within that domain, the theorem states that the absolute matter-sector zero-point coefficient is a coordinate on a renormalization convention, not an invariant prediction. The theorem is intentionally local to the homogeneous metric-proportional sector and does not classify the full renormalized stress tensor in arbitrary curved backgrounds.

Consequently, the paper removes only the zero-point prediction formulation of the cosmological-constant problem. It leaves intact the naturalness, radiative-stability, and matching questions for the measured renormalized parameter Λ_{eff} . This scope restriction is part of the result, not a concession after the fact.

2 Predictions and parameters in a renormalized theory

We use signature $(-, +, +, +)$ and units $\hbar = c = 1$. A renormalized QFT contains both invariant predictions and scheme-dependent coordinates on a space of descriptions. The distinction matters. A bare coupling, cutoff-dependent loop contribution, subtraction constant, or finite counterterm convention can be indispensable for calculation without being a directly predicted observable number.

A physical prediction is normally a relation among quantities after empirical normalization conditions have fixed the necessary renormalized inputs. A renormalized parameter may be physical in the sense that it is measured and then used in predictions. But its numerical value is not predicted by the low-energy theory merely because the parameter appears in the Lagrangian. Charge renormalization is the standard example: loop corrections do not by themselves predict the measured electron charge; once the charge is fixed by a normalization condition, the theory predicts scattering amplitudes, running, and relations among observables.

The same distinction is essential for vacuum energy. The question is not whether a cosmological term is physically important. It is. The question is whether the absolute matter-sector zero-point

term has a standalone invariant value before a gravitational normalization condition fixes the total cosmological coefficient.

Definition 1 (Finite-renormalization equivalence). Let $\mathfrak{R}$ denote finite transformations of the renormalized description that preserve locality, covariance, conservation of the stress tensor, and all empirical normalization conditions already imposed on the non-vacuum matter observables under discussion. Two descriptions D and D' are equivalent, written $D \sim_{\mathfrak{R}} D'$, when they are related by such a finite transformation.

Definition 2 (Standalone prediction). A quantity is treated here as a standalone prediction only if its value is invariant under admissible scheme transformations once empirical normalization conditions are fixed. More explicitly, a scalar assignment $X(D)$ is a standalone prediction of the renormalized theory only if its finite value is fixed without using an empirical normalization condition for X itself and if it descends to the quotient by $\sim_{\mathfrak{R}}$:

$$X(D) = X(D') \quad \text{whenever} \quad D \sim_{\mathfrak{R}} D'. \quad (2)$$

A non-invariant scalar may still be a useful scheme coordinate or a renormalized parameter. It is not, by itself, a predicted observable number.

This criterion avoids an overstatement. It is not being claimed that every numerical parameter appearing in physics must be invariant as a standalone number. Renormalized parameters transform with conventions; physical predictions remain invariant when parameters and computed quantities are transformed together. The claim is only that a regulator-dependent zero-point contribution cannot be compared directly with observation as if it were an invariant prediction.

3 Stress-tensor renormalization and the homogeneous vacuum term

The stress-energy tensor is a composite operator. In curved spacetime, its renormalized definition is constrained by locality, covariance, conservation, and appropriate state-dependence requirements, but these constraints do not fix it uniquely [6,8–11]. The remaining ambiguity is local and conserved. It is generated by finite changes in the local gravitational counterterms.

A representative low-energy gravitational action has the form

$$S_{\text{grav}} = \int d^4x \sqrt{-g} \left[-\frac{\Lambda_{\text{R}}}{8\pi G_{\text{R}}} + \frac{R}{16\pi G_{\text{R}}} + a_{\text{R}} R^2 + b_{\text{R}} R_{\alpha\beta} R^{\alpha\beta} + c_{\text{R}} R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} + \dots \right]. \quad (3)$$

Finite redefinitions of these coefficients induce corresponding finite conserved shifts of the renormalized stress tensor. Schematically,

$$\langle T_{\mu\nu} \rangle_{\text{R}} \longrightarrow \langle T_{\mu\nu} \rangle_{\text{R}} - \delta\rho g_{\mu\nu} + \delta A G_{\mu\nu} + \delta B H_{\mu\nu}^{(1)} + \delta C H_{\mu\nu}^{(2)} + \dots, \quad (4)$$

where $H_{\mu\nu}^{(1)}$ and $H_{\mu\nu}^{(2)}$ denote the conserved tensors obtained by varying curvature-squared terms. These curvature-dependent terms matter for general curved backgrounds, trace anomalies, and higher-derivative gravitational response. They are not the target of the present theorem.

The target is the homogeneous metric-proportional component. For a Lorentz-invariant vacuum sector on a fixed background class, write

$$\langle T_{\mu\nu} \rangle_{0,\text{R}} = -\rho_{\text{R}} g_{\mu\nu}. \quad (5)$$

The finite transformation in Eq. (4) includes

$$\rho_R \longrightarrow \rho_R + \delta\rho. \quad (6)$$

This shift leaves the non-vacuum matter normalization conditions unchanged. It changes only the convention-dependent split between an additive matter-sector vacuum term and the gravitational cosmological counterterm.

The cutoff expression in Eq. (1) should be understood in this context. Its divergent and finite pieces are local contributions to the renormalization of gravitational couplings. In flat space the quartic divergence is the metric-proportional term; in curved space other terms renormalize Newton's constant and higher-curvature couplings. None of these raw regulated pieces is a standalone observable.

Remark 1 (Trace anomalies and curvature response). The theorem below does not deny curvature-dependent quantum effects such as trace anomalies. Such effects belong to the local curvature-response part of Eq. (4). They are distinct from an absolute homogeneous vacuum-energy offset in a fixed vacuum sector. The argument concerns only the latter.

4 Classification under the homogeneous vacuum shift

We now isolate the minimal transformation responsible for the zero-point prediction failure.

Definition 3 (Homogeneous vacuum-energy sector). Let $\mathcal{V}_0$ be the sector of scalar assignments depending only on the coefficient ρ_R of the metric-proportional homogeneous vacuum stress tensor in Eq. (5), on a fixed background class, with curvature-dependent counterterms, boundary/material counterterms, and non-vacuum matter normalization conditions fixed independently. The finite transformation group acting on this sector is

$$\mathcal{G}_0 : \quad \rho_R \mapsto \rho_R + \delta\rho, \quad (7)$$

for arbitrary finite $\delta\rho$.

This definition is deliberately narrow. It does not range over every possible local curvature observable in QFT on curved spacetime. It isolates the single homogeneous additive ambiguity that appears in the standard zero-point comparison.

Theorem 1 (No standalone absolute zero-point prediction). *No nonconstant scalar depending only on ρ_R in $\mathcal{V}_0$ is invariant under $\mathcal{G}_0$. Consequently, the absolute homogeneous matter-sector vacuum energy is not a standalone prediction of QFT.*

Proof. Let $X(\rho_R)$ be a scalar assignment in $\mathcal{V}_0$ that is invariant under the finite shift in Eq. (7). Then

$$X(\rho) = X(\rho + \delta\rho) \quad \text{for every finite } \delta\rho \in \mathbb{R}. \quad (8)$$

The translation orbit of any ρ under $\mathcal{G}_0$ is all of $\mathbb{R}$. Hence X is constant on the quotient. A constant assignment carries no information about the absolute value of ρ_R . Therefore the absolute homogeneous vacuum-energy coefficient has no nonconstant invariant content in $\mathcal{V}_0$. $\square$

The content of the result is not algebraic difficulty but scope restriction: it identifies the maximal class of quantities for which the zero-point construction could in principle yield invariant predictions, and shows that this class excludes absolute vacuum energy. The result does not assert that vacuum contributions to the gravitational effective action vanish; it asserts that the zero-point summation procedure does not define a standalone predictive map to that parameter.

Theorem 2 (Exhaustion within the restricted homogeneous shift sector). *Within the restricted homogeneous metric-proportional shift sector $\mathcal{V}_0$, every invariant use of vacuum-energy data falls into one of the following classes:*

- (i) *boundary-, material-, phase-, or configuration-conditioned differences for which a common additive offset cancels;*
- (ii) *derivatives or response quantities built from such differences;*
- (iii) *the gravitationally renormalized effective cosmological constant, which is invariant only as a total coefficient and not as a unique split into matter vacuum energy plus a geometric cosmological term.*

There is no fourth invariant corresponding to the absolute zero-point energy of a single fixed vacuum sector.

Proof. The action of $\mathcal{G}_0$ is the additive translation action on the coordinate ρ_R . For a single fixed homogeneous vacuum, Theorem 1 shows that the quotient is trivial: no nonconstant invariant of the absolute coordinate remains.

For a family of physically specified configurations labelled by a , the same common homogeneous offset acts as

$$E_R(a) \mapsto E_R(a) + C, \quad (9)$$

where C is independent of the compared configuration after the same background and apparatus convention has been used. Invariants under this common shift are functions of differences such as $E_R(a) - E_R(a_0)$, derivatives such as $\partial E_R / \partial a$, or dynamical transition quantities depending on energy differences between related configurations. These give classes (i) and (ii).

When semiclassical gravity is included, the same additive matter-sector shift is absorbed into the cosmological term. As shown explicitly in Section 5, curvature observations determine only the total coefficient Λ_{eff} . This gives class (iii). These alternatives exhaust the invariant structures generated by the homogeneous additive data specified in $\mathcal{V}_0$. $\square$

Remark 2 (Restriction of the exhaustion claim). Theorem 2 is intentionally local to the homogeneous metric-proportional sector. It does not classify the full renormalized stress tensor in arbitrary curved backgrounds, where curvature-dependent conserved tensors, anomalies, state-dependent terms, and boundary/material renormalizations must be handled separately. Its target is only the additive homogeneous vacuum-energy coordinate used in the standard zero-point comparison.

Corollary 1 (The zero-point comparison is not a prediction failure). *The comparison between Eq. (1) and the observed cosmological constant is not a comparison between a standalone QFT prediction and experiment. It compares a regulator- and subtraction-dependent intermediate quantity with a renormalized empirical gravitational parameter.*

Proof. Equation (1) changes under regulator and finite subtraction convention. Theorems 1 and 2 show, within the restricted homogeneous metric-proportional sector, that the corresponding finite absolute coefficient is not invariant under the relevant finite renormalization. Therefore it is not a standalone prediction. The observed cosmological constant fixes the total renormalized gravitational coefficient, not the matter-sector zero-point term separately. $\square$

5 Semiclassical gravity gives non-identifiability, not a separate measurement

A common objection is that vacuum energy becomes physical because gravity couples to the stress tensor. This objection correctly identifies why the cosmological constant is physically important. It does not establish the separate observability of the matter-sector vacuum-energy split.

Write the semiclassical field equation as

$$G_{\mu\nu} + \Lambda_R g_{\mu\nu} = 8\pi G_R \langle T_{\mu\nu} \rangle_R. \quad (10)$$

For a decomposition

$$\langle T_{\mu\nu} \rangle_R = -\rho_R g_{\mu\nu} + \langle T_{\mu\nu} \rangle_{\text{excit}}, \quad (11)$$

Eq. (10) becomes

$$G_{\mu\nu} + (\Lambda_R + 8\pi G_R \rho_R) g_{\mu\nu} = 8\pi G_R \langle T_{\mu\nu} \rangle_{\text{excit}}. \quad (12)$$

Thus, with the present sign convention,

$$\Lambda_{\text{eff}} = \Lambda_R + 8\pi G_R \rho_R. \quad (13)$$

Lemma 1 (Degeneracy of the split). *The transformation*

$$\rho_R \mapsto \rho_R + \delta\rho, \quad \Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho \quad (14)$$

leaves Eq. (12) invariant.

Proof. Substituting Eq. (14) into Eq. (13) gives

$$\Lambda_R - 8\pi G_R \delta\rho + 8\pi G_R (\rho_R + \delta\rho) = \Lambda_R + 8\pi G_R \rho_R. \quad (15)$$

All curvature predictions depending on Eq. (12) are unchanged. $\square$

Proposition 1 (Non-identifiability in semiclassical gravity). *Within semiclassical gravity, no curvature observation determines ρ_R and Λ_R separately along the orbit in Eq. (14). Observations determine Λ_{eff} , not a unique decomposition into matter-sector vacuum energy and a geometric cosmological term.*

Proof. Two descriptions related by Eq. (14) produce the same Λ_{eff} and hence the same metric-proportional contribution to the field equation. Since the equations governing curvature and matter excitations are identical after the simultaneous finite redefinition, any curvature observable constructed from the semiclassical solution is identical in the two descriptions. Therefore no such observation separates the two coordinates of the split. $\square$

The conclusion is not that the cosmological constant is unobservable. It is that the observed quantity is the total renormalized coefficient Λ_{eff} . Coupling to gravity turns the homogeneous vacuum shift into cosmological-constant renormalization. It does not promote the matter-sector zero-point coordinate into an independently measured source.

6 Relational survivors: Casimir forces and phase differences

The exhaustion theorem says that relational quantities can remain physical. Two standard cases are Casimir forces and dynamical energy differences between physical phases. Both are sometimes used to argue that absolute vacuum energy is observable. Neither gives that conclusion.

6.1 Casimir forces

For ideal conducting parallel plates separated by distance a , the conventional zero-temperature result is

$$\frac{F}{A} = -\frac{\pi^2}{240} \frac{1}{a^4}, \quad (16)$$

with the factor $\hbar c$ restored in ordinary units. The physical result is a force or pressure difference. It can be represented as a derivative of a renormalized energy difference,

$$E_{\text{Cas}}(a) = E_{\text{R}}(a) - E_{\text{R}}(\infty), \quad F_{\text{Cas}}(a) = -\frac{\partial E_{\text{Cas}}}{\partial a}. \quad (17)$$

The relevant claim is not that all formal divergences in idealized Casimir calculations cancel automatically. Real calculations require material response, geometry, finite conductivity, temperature, and local boundary or surface counterterms. The claim is narrower: the measured force is a response of a specified material-boundary system and is invariant under a uniform shift of the global vacuum-energy zero. The argument does not depend on idealized perfect-conductor calculations; it applies to fully renormalized Casimir forces in realistic material systems, where the measured force remains invariant under shifts of the global vacuum-energy zero.

This point is independent of whether one prefers a zero-point-energy representation, a stress-tensor representation, or a scattering/charge-current representation of the Casimir effect. Jaffe emphasized that Casimir forces can be formulated without taking zero-point energies as directly real physical stuff; they are ordinary relativistic quantum forces between material sources [15–18]. This supports rather than weakens the present classification.

Proposition 2 (Casimir invariance). *Casimir forces are invariant under the homogeneous additive finite renormalization that prevents absolute zero-point energy from being a standalone prediction.*

Proof. Under a uniform shift $E_{\text{R}}(a) \mapsto E_{\text{R}}(a) + C$ with C independent of the compared plate separation,

$$(E_{\text{R}}(a) + C) - (E_{\text{R}}(\infty) + C) = E_{\text{R}}(a) - E_{\text{R}}(\infty). \quad (18)$$

The derivative with respect to a is also unchanged. Boundary and material counterterms can affect the definition of the apparatus and its self-energies, but the measured force is a configuration response after that apparatus has been fixed. It is not a measurement of an absolute homogeneous vacuum offset. $\square$

6.2 Phase transitions and dynamically connected vacua

The argument also does not deny the physical relevance of energy differences between distinct phases, metastable vacua, or dynamically connected configurations. Such differences can enter tunneling rates, domain-wall dynamics, bubble nucleation, and cosmological histories [19, 20]. They are relational quantities: they compare two physical configurations under a common convention. A common additive offset of all vacuum energies does not change the transition rate, wall tension, or energy difference once the gravitational background and matching conditions are specified.

Thus phase differences belong to class (i) of Theorem 2. They can be physically meaningful without making the absolute zero-point energy of a single isolated vacuum sector observable. In a gravitational setting their net effect is encoded in the effective cosmological term appropriate to each branch or phase, not in an independently measurable universal zero-point baseline.

7 What remains: the naturalness and matching problem

The result removes one formulation of the cosmological constant problem: the claim that QFT has produced a definite numerical prediction for Λ_{eff} by summing zero-point modes and has thereby failed by many orders of magnitude. It does not remove the smallness problem for the measured renormalized cosmological constant.

In effective field theory, the cosmological constant is a relevant coupling in the gravitational action. Heavy fields and symmetry breaking sectors contribute to threshold matching for this coupling. Unless protected by a symmetry or mechanism, one expects radiative corrections of order the heavy scales available in the effective description. Explaining why the measured value of Λ_{eff} is so small, and why it is radiatively stable under changes of EFT description, is the legitimate technical-naturalness problem emphasized in EFT analyses [4]. The present paper does not solve that problem.

The point is instead diagnostic. A solution to the remaining problem must explain the renormalized gravitational coefficient or the boundary condition fixing it. It need not cancel a unique, scheme-independent matter-sector number equal to Eq. (1), because no such number exists within the restricted homogeneous shift sector. Supersymmetry, adjustment fields, sequestering, anthropic selection, or ultraviolet completions can be assessed as proposed explanations for Λ_{eff} or its matching behavior. They should not be framed as repairing a failed standalone QFT prediction of absolute zero-point energy.

Corollary 2 (No scheme-independent cancellation target). *There is no scheme-independent matter-sector absolute vacuum-energy number that a cancellation mechanism must cancel in order to repair the naive zero-point discrepancy. The invariant target is the measured or theoretically derived value of Λ_{eff} .*

Proof. By Theorem 1, the absolute homogeneous matter-sector vacuum coefficient changes under an allowed finite renormalization. By Lemma 1, the same change is compensated by a shift of Λ_R without altering Λ_{eff} . A cancellation of the split-dependent matter coordinate is therefore convention-dependent. The invariant quantity available to be matched or predicted is the total effective cosmological constant. $\square$

8 Objections and replies

Objection 1: the argument is already known. The component facts are known: zero-point sums are regulator-dependent, stress tensors admit finite renormalization ambiguities, and semiclassical gravity observes a combined cosmological parameter. The contribution here is not a new loop calculation and not a claim of priority for those facts. It is the explicit classification of the standard zero-point comparison under the homogeneous metric-proportional finite renormalization. In particular, the manuscript identifies the comparison itself as the failed prediction-forming step: the zero-point expression is not merely numerically wrong, but not a standalone prediction of the relevant renormalized theory.

Objection 2: gravity makes vacuum energy physical. Gravity makes the total cosmological term physical. It does not make the split in Eq. (13) separately identifiable. Two descriptions connected by Eq. (14) have the same curvature predictions. Therefore gravitational observation fixes Λ_{eff} , not ρ_R independently.

Objection 3: observables need not be invariant numbers. Correct: renormalized parameters can change their numerical coordinates under changes of convention, provided all predictions are transformed consistently. That is why the paper uses the narrower notion of a standalone prediction. The zero-point term can be a scheme coordinate in a calculation. It is not a predicted observable number until a normalization condition fixes the total parameter in which it appears.

Objection 4: curved-spacetime stress tensors have more ambiguities than a constant shift. Also correct. Equation (4) includes curvature-dependent conserved tensors. The theorem is explicitly restricted to $\mathcal{V}_0$, the homogeneous metric-proportional vacuum sector relevant to the zero-point comparison. Trace anomalies and curvature response effects are outside that restricted target and are not denied.

Objection 5: Casimir forces show vacuum energy is real. Casimir forces show that boundary-conditioned quantum response quantities are real. They do not measure the absolute homogeneous vacuum-energy offset. The force is invariant under a uniform shift of the energy zero and can be formulated without assigning direct ontological status to a bulk zero-point density [18].

9 Conclusion

The cutoff zero-point energy of quantum fields is often treated as an absolute matter-sector quantity whose numerical value conflicts with cosmology. That treatment is not supported by the structure of the renormalized theory. The stress-energy tensor admits a finite metric-proportional renormalization. This transformation shifts the homogeneous vacuum-energy coefficient while preserving the physical matter theory. In semiclassical gravity the same shift is exactly degenerate with a redefinition of the cosmological constant. No curvature observation separates the two.

The invariant remnants are relational differences, response quantities, phase or boundary comparisons, and the total renormalized cosmological coefficient. Casimir forces and phase-transition energy differences remain physical precisely because they are relational. They do not license the inference to an observable absolute zero-point density.

Therefore the standard comparison between a cutoff zero-point sum and the observed cosmological constant is not a standalone failed prediction of QFT. It is a comparison between a regulator-dependent intermediate quantity and an empirical renormalized gravitational parameter. The zero-point prediction formulation is removed; the measured smallness of Λ_{eff} remains a real naturalness and matching problem. What fails is the stronger claim that quantum field theory independently predicts the absolute matter-sector vacuum energy by summing zero-point modes.

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